

LIQUID - GAS DIFFUSIVITY

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Objective:

The aim of this experiment is to determine the diffusion coefficient of acetone (liquid) in air (gas).

Theory and Background:

Diffusivity tells us how fast a substance spreads or mixes into another due to a difference in concentration. The molecules move from high concentration to low concentration. In this experiment, we studied how acetone, a highly volatile liquid, evaporates into the air through diffusion.

To do this, acetone was placed in a glass test tube. A steady stream of air was directed over the top of the tube to maintain a nearly constant concentration of acetone vapor in the air. This ensured that the evaporation rate depended only on the diffusion process.

Winkelman's method is a method used to measure the diffusivity of a liquid in air. It involves monitoring the rate at which the liquid level drops due to evaporation under controlled conditions. At constant temperature there will be no eddy currents in the vertical tube and mass transfer from the surface of liquid in the test tube will take place only by diffusion. The rate of mass transfer can be determined from the rate of fall of liquid level in the test tube. The method assumes that the evaporation follows Fick's law of diffusion, which describes how molecules move from high to low concentration areas.

Fick's first law states that diffusion is driven by the concentration gradient of mass transfer. The diffusivity coefficient (D) indicates the rate, and it depends on the pressure, temperature, and other properties of the species considered like volatility.

Fick's law of diffusion is stated as: $N_A = -D_{AB} \frac{dC_A}{dz} + x_A(N_A + N_B)$

Where: N_A = mass flux of A

D_{AB} = Diffusivity coefficient

C_A = concentration of component A

z = height of the air column or the diffusion length

$$N_A = -D_{AB} \frac{dC_A}{dz} + x_A(N_A + N_B)$$

$$N_B = 0 \Rightarrow N_A = -D_{AB} \frac{dC_A}{dz} + x_A N_A$$

($N_B = 0$) as B is non-diffusing

$$C_A = \frac{p_A}{RT}$$

$$N_A = \frac{D_{AB}}{z} C_T \frac{\Delta C_A}{CBlm}$$

$$\Delta C_A = \frac{C_{A1} - C_{A2}}{CBlm}, C_{A2} = 0, CBlm = \frac{C_{B2} - C_{B1}}{\ln\left(\frac{C_{B2}}{C_{B1}}\right)}$$

$$N_A = \frac{D_{AB}}{z} \frac{P_T}{RT} \frac{\Delta C_A}{CBlm}$$

$$N_A = C_A \frac{dz}{dt} = \frac{\rho_A}{M_A} \frac{dz}{dt}$$

$$\frac{\rho_A}{M_A} \frac{dz}{dt} = \frac{D_{AB}}{z} C_T \frac{\Delta C_A}{CBlm}$$

$$\int_{z_0}^z z dz = D_{AB} \frac{M_A}{\rho_A} C_T \frac{\Delta C_A}{CBlm} \int_0^t dt$$

$$\left[\frac{z^2}{2} \right]_{z_0}^z = D_{AB} \frac{M_A}{\rho_A} C_T \frac{\Delta C_A}{CBlm} \int_0^t dt$$

$$\frac{[z^2 - z_0^2]}{2} = D_{AB} \frac{M_A}{\rho_A} C_T \frac{\Delta C_A}{CBlm} t$$

$$\frac{(z-z_0)(z+z_0)}{2} = D_{AB} \frac{M_A}{\rho_A} C_T \frac{C_{A1} - C_{A2}}{CBlm} t, C_{A2} = 0$$

$$\frac{\rho_A \frac{C_{B2} - C_{B1}}{\ln\left(\frac{C_{B2}}{C_{B1}}\right)} (z-z_0)}{2 D_{AB} C_T C_{A1} M_A} = \frac{t}{z+z_0}$$

M_A = Molecular weight, ρ_A = Density, C_T = Total concentration

$CBlm$ is the log mean concentration.

Slope of plot of $\frac{t}{z+z_0}$ vs $(z - z_0)$ is used to calculate diffusivity

Schematic of the experiment:

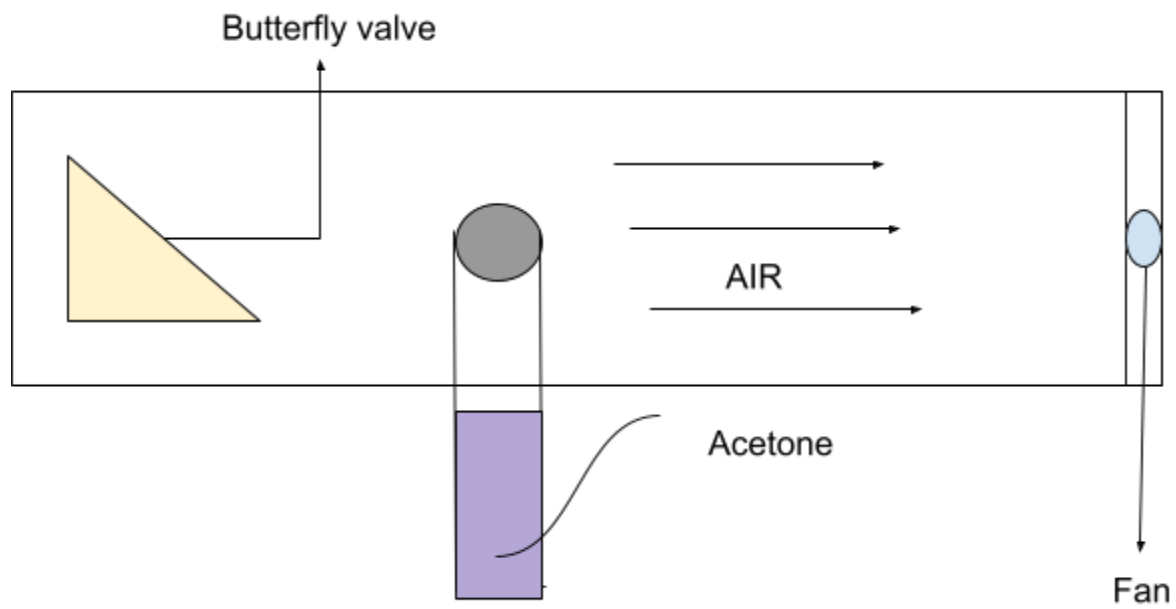


Figure 0: Schematic diagram of the experiment

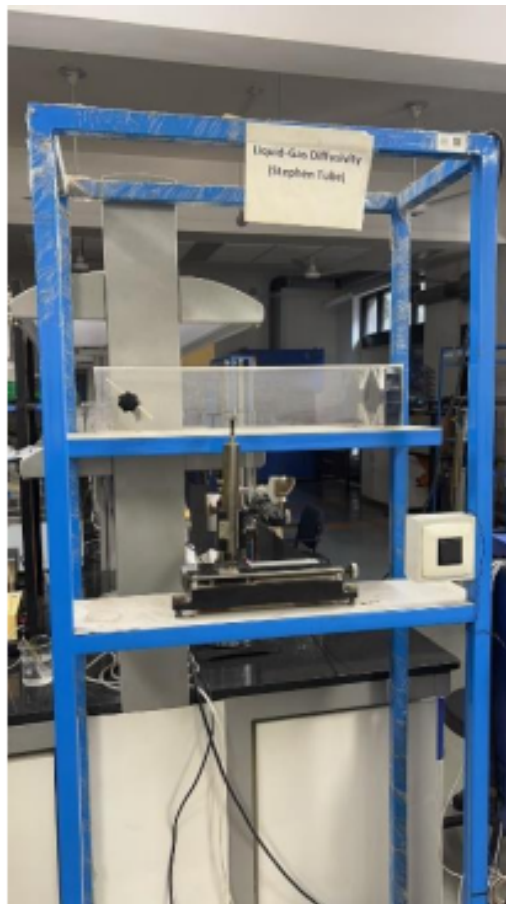


Figure 1: Figure showing the experimental setup

Apparatus and Procedure:

Apparatus:

1. Acetone
2. Test tube (approximately 70% filled with acetone)
3. Duct with air current system
4. Butterfly valve for air velocity control
5. Travelling microscope
6. Stopwatch or timer
7. Measuring scale
8. Fan (to maintain constant air current)

Procedure:

1. Take a clean test tube and fill it with acetone. The acetone should be filled to about 70% of the maximum level.
2. Attach the filled test tube to the bottom of the duct. Mark and note down the initial height of the liquid.
3. Switch on the fan and maintain the fan at a constant speed.
4. Adjust the butterfly valve to have maximum velocity of air current through the duct.
5. Calculate the height of the liquid in the test tube using a travelling microscope after every 25 minutes.
6. Take 5 readings.

Known parameters:

- *Initial height of Acetone in test tube: 8.7 cm*
- *Height of the test tube: 11 cm*
- *Initial height of air column: 2.3 cm*

Readings:

TABLE 1: TABLE CONTAINING EXPERIMENTAL DATA

Time (s)	Height of Acetone (cm)	Height of Air column (cm)
0	8.7	2.3
1500	8.6	2.4
3000	8.5	2.5
4500	8.4	2.6
6000	8.3	2.7
7500	8.3	2.7

TABLE 2: TABLE CONTAINING CALCULATED DATA

Time (s)	Height of Acetone (cm)	Height of Air column (cm)	$(Z - Z_0)$ (cm)	$\frac{t}{Z + z_0} \left(\frac{s}{cm} \right)$
0	8.7	2.3	0	0
1500	8.6	2.4	0.1	319.1489362
3000	8.5	2.5	0.2	625
4500	8.4	2.6	0.3	918.3673469
6000	8.3	2.7	0.4	1200
7500	8.3	2.7	0.4	1500

Results and discussion:

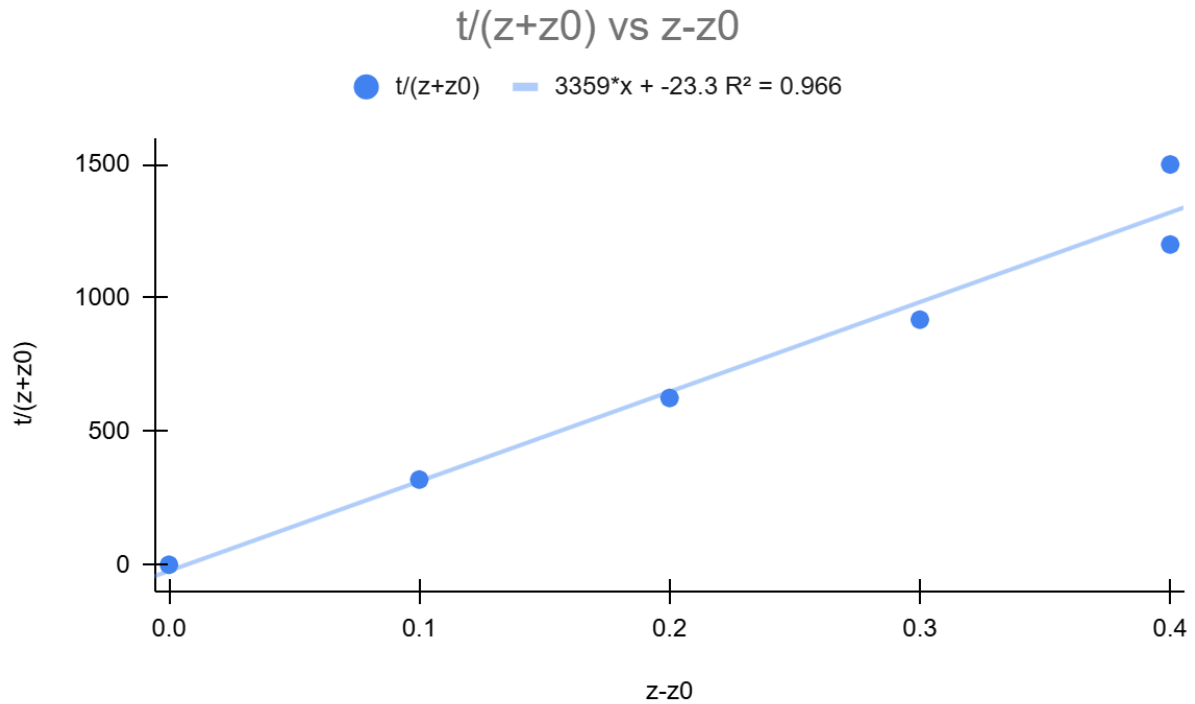


Figure 2: Figure showing the plot between $\frac{t}{z + z_0}$ vs $(z - z_0)$

Slope of the graph is $3359 \frac{s}{cm^2}$

From figure 1, we can see that there is a linear relationship between $\frac{t}{z + z_0}$ and $(z - z_0)$, which confirms the validity of Fick's law.

In this experiment the Acetone diffuses in air and not the other way because:

1. Acetone is More Volatile

- Acetone evaporates easily because its molecules have high kinetic energy. This allows them to break free from the liquid surface and enter the air.
- On the other hand, air (mainly nitrogen and oxygen) has very low solubility in acetone, so it doesn't dissolve into the liquid in any significant amount.

2. Concentration Gradient Drives Diffusion

- Diffusion always happens from a region of high concentration to low concentration. In this case, acetone has a high concentration in the liquid phase, but a much lower concentration in the air.
- This imbalance pushes acetone molecules upward, while air molecules don't have the same driving force to move into the liquid.

3. Continuous Airflow Keeps Acetone Evaporating

- A stream of air is directed over the tube's opening, ensuring that acetone vapor doesn't build up in the surrounding air.
- This prevents equilibrium from being reached, meaning acetone keeps evaporating while air remains unchanged.

4. Nature of the Liquid-Air Interface

- Acetone molecules at the surface experience weaker attraction to the liquid than those inside, making it easier for them to escape into the air.
- In contrast, air molecules don't have a strong tendency to dissolve into acetone, so their diffusion into the liquid is minimal.

Calculations:

$$\text{Density of the liquid } (\rho) = 788 \frac{\text{g}}{\text{L}} = 788 \times 10^{-3} \frac{\text{kg}}{\text{L}}$$

$$\text{Temperature} = 26^\circ \text{C}$$

$$M = 60 \frac{\text{g}}{\text{mol}}$$

$$\text{Antoine Equation: } \ln(P^{\text{sat}}) = A - \frac{B}{T + C}$$

$$A = 4.4245, B = 1312.253, C = 32.445, T = 273 + 26 \text{ K} = 299 \text{ K}$$

$$P^{\text{sat}} = \exp\left(4.4245 - \frac{1312.253}{299 + 32.445}\right) = 1.59 \text{ atm}$$

$$C_T = \frac{1}{v} \left(\frac{273}{T} \right) = \frac{1 \cdot 273}{22.4 \cdot 299} = 0.048 \frac{\text{mol}}{\text{L}}$$

$$C_A = P^{\text{sat}} \times \frac{C_T}{P_T} = 0.07632 \frac{\text{mol}}{\text{L}}$$

$$CBlm = \frac{C_{B2} - C_{B1}}{\ln\left(\frac{C_{B2}}{C_{B1}}\right)} = \frac{0.0408 - 0.016}{\ln\left(\frac{0.0408}{0.016}\right)} = 0.0264 \frac{\text{mol}}{\text{L}}$$

$$\text{Slope of the graph is } 3359 \frac{\text{s}}{\text{cm}^2}$$

$$D_{AB} = \frac{\rho CBlm}{2 \cdot M \cdot \text{slope} \cdot C_A \cdot C_T} = 1.4 \times 10^{-5} \frac{\text{m}^2}{\text{s}}$$

$$\text{Diffusivity of acetone in air calculated in this experiment} = 1.4 \times 10^{-5} \frac{\text{m}^2}{\text{s}}$$

$$\text{Theoretical diffusivity of acetone in air} = 1.68 \times 10^{-5} \frac{\text{m}^2}{\text{s}}$$

$$\% \text{ error} = \frac{\text{Theoretical value} - \text{experimental value}}{\text{Theoretical value}} = \frac{1.68 - 1.4}{1.68} \times \frac{10^{-5}}{10^{-5}} \times 100 = 16.66 \%$$

Conclusion:

This experiment successfully determined the diffusivity of acetone in air using Winkelmann's method and compared it with the theoretical value. The method is based on Fick's first law of diffusion, and a graph of $\frac{t}{z + z_0}$ and $(z - z_0)$ was plotted to calculate the diffusion coefficient.

The experimental value of Diffusivity of acetone in air is $1.4 \times 10^{-5} \frac{\text{m}^2}{\text{s}}$ and the theoretical value of diffusivity of acetone in air $= 1.68 \times 10^{-5} \frac{\text{m}^2}{\text{s}}$. The experimental value showed a 16.66% deviation from the theoretical value. The deviation of 16.66% can be justified because experimental conditions don't match theoretical models perfectly. Theoretical diffusivity calculations assume ideal conditions, while in practice, minor fluctuations in air movement, liquid level readings, and environmental factors contribute to measurement uncertainty. Despite these variations, the experiment effectively validated Winkelmann's method and demonstrated the practical application of mass transfer principles in diffusion studies.

This study holds significance in various industries, including chemical engineering, environmental science, and industrial solvent recovery, where understanding diffusion behavior is crucial for process optimization, air quality control, and equipment design.

Sources of errors:

- There can be slight error in taking the readings as we have to note the lower meniscus.
- Improper adjustment of the butterfly valve could affect the diffusion rate.

- We have assume $N_B = 0$ which can not be completely true.
- We have applied the quasi steady state condition for diffusion where thickness of the gas layer above the liquid surface is taken into consideration while the rate of diffusion is independent of the amount of liquid taken in the test tube.

Precautions:

- Wear gloves while handling liquids.
- Use pure acetone.
- The test tube and other instruments should be cleaned before use.
- It is important to take accurate height measurements carefully and without error.

References:

- [1] D. W. Green and R. H. Perry, *Perry's Chemical Engineers' Handbook, Eighth Edition*. McGraw-Hill Education, 2008. Available: [Link](#)
- [2] 'Lab Manual' provided in the classroom.
- [3] K. Murthy, S. Tadepalli, and A. Alfozan, "MOLECULAR DIFFUSION IN GASES," Jan. 2023. Available: [Link](#)
- [4] "Mass Transfer Operations Robert Treybal" *Internet Archive*, 2023. [Link](#)